3-24-26 (WEEK 5)

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EXAMPLE (COUPON COLLECTOR PROBLEM)

Every purchase we make, we get a coupon, which is uniformly randomly chosen from one of n types, indep of previous coupons

Goal: Get a full set of coupons

Random variable: $T = \#$purchases before we have a full set

Question: What is the distribution of T, i.e. for $k \in \mathbb{N}$, what is $p(k) = \mathbb{P}(T=k)$?

Solution

Sample space: $S = \{(s_1, s_2, s_3, \dots) \mid s_i \in \{1, 2, \dots, n\}\}$ (type of ith coupon)

Notice, $\mathbb{P}(\{s_i = j\}) = \frac{1}{n}$ ($\because$ coupons: unif rand)

By independence, $\mathbb{P}(\{s_i = j_i \mid 1 \le i \le k\}) = \frac{1}{n^k}$ $\forall j_i \in [n]$ (☆)

Random Variable: $T((s_1, s_2, \dots)) = \min\{k : (s_1, \dots, s_k) \text{ contains types } 1 \text{ to } n\}$ (if such a k exists)

Notice, $\mathbb{P}(T=k) = 0$ $\forall k < n$ ($\because$ we need $\ge n$ coupons to have all n types)

$\mathbb{P}(T=n)$: Let E be the event that $T=n$

By ☆, the first n coupons are unif distributed over all possible seq)

$\#$possible seq $= n^n$ (product rule)

$\#$good seq $= n!$ (perm of n types)

$\therefore \mathbb{P}(T=n) = \mathbb{P}(E) = \frac{n!}{n^n} \underset{\text{Stirling approx}}{\approx} \left(\frac{1}{e}\right)^n \xrightarrow{n \to \infty} 0$

Moreover, $\mathbb{P}(\text{no complete set of coupons out of } mn \text{ purchases}) \le \left(1 - \frac{n!}{n^n}\right)^m \xrightarrow{m \to \infty} 0$

For $k > n$, what is $\mathbb{P}(T=k)$?

We can start by considering $\mathbb{P}(T>k)$

Let $E_i^{(k)} = \{\text{coupon of type } i \text{ didn't appear in first } k \text{ coupons}\}$

Then, $\{T>k\} = \bigcup_{i=1}^n E_i^{(k)}$

By indep, $\mathbb{P}(E_i^{(k)}) = \left(\frac{n-1}{n}\right)^k$ (coupon 1: type not $i \to \frac{n-1}{n}$; coupon 2: $\frac{n-1}{n}$; ...)

Side note: $E_1^{(k)}, \dots, E_n^{(k)}$: not indep ($\because \mathbb{P}(\bigcap_{i=1}^n E_i^{(k)}) = 0$)

By incl-excl,

$$\mathbb{P}\left(\bigcup_{i=1}^n E_i^{(k)}\right) = \sum_{\substack{I \subseteq [n] \\ I \neq \emptyset}} (-1)^{|I|+1} \underbrace{\mathbb{P}\left(\bigcap_{i \in I} E_i^{(k)}\right)}_{\text{By indep, equals } \left(\frac{n-|I|}{n}\right)^k}$$

$$= \sum_{\substack{I \subseteq [n] \\ I \neq \emptyset}} (-1)^{|I|+1} \left(\frac{n-|I|}{n}\right)^k$$

$$= \sum_{r=1}^{n-1} \binom{n}{r} (-1)^{r+1} \left(\frac{n-r}{n}\right)^k = \mathbb{P}(T>k)$$

($r = |I|$; choose I of size r)

As $\{T>k-1\} = \{T=k\} \sqcup \{T>k\}$, thus $\mathbb{P}(T=k) = \mathbb{P}(T>k-1) - \mathbb{P}(T>k)$

Simplifying this, we get

$$\mathbb{P}(T=k) = \sum_{r=1}^{n-1} \binom{n}{r} (-1)^{r+1} \left(\frac{n-r}{n}\right)^{k-1} - \sum_{r=1}^{n-1} \binom{n}{r} (-1)^{r+1} \left(\frac{n-r}{n}\right)^k = \sum_{r=1}^{n-1} \binom{n}{r} (-1)^{r+1} \left(\frac{n-r}{n}\right)^{k-1} \cdot \frac{r}{n}$$

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EXPECTED VALUE

DEFINITION

Let $(S, \mathbb{P})$ be a probability space and let $X: S \to \mathbb{R}$ be a discrete random variable

Then, the expected value of X is $\mathbb{E}[X] = \sum_{x_i} x_i p(x_i) = \sum_{x_i} x_i \mathbb{P}(X = x_i)$

INTERPRETATION

Expected value is what we expect to see on average if we repeat the random experiment many times.

Proof

Suppose we repeat N times, and let $n_i := \#$ times we see $X = x_i$

Then, the average value of $X = \frac{\sum_i x_i n_i}{N} = \sum_i x_i \frac{n_i}{N} \to \sum_i x_i \mathbb{P}(X = x_i)$ $\square$

EXAMPLE

Two MCQ, each with 4 options.

If we choose answers unifly random, how many do we expect to get correct?

Solution

Let $X = \#$ correct ans

x_i	$\mathbb{P}(X = x_i)$
0	$(\frac{3}{4})^2 = \frac{9}{16}$
1	$2(\frac{3}{4})(\frac{1}{4}) = \frac{3}{8}$
2	$(\frac{1}{4})^2 = \frac{1}{16}$

$\therefore \mathbb{E}[X] = \sum_i x_i \mathbb{P}(X = x_i) = 0(\frac{9}{16}) + 1(\frac{3}{8}) + 2(\frac{1}{16}) = \frac{1}{2}$

Note that this is not the "most common" x_i nor is it even attainable!

CLAIM

$\mathbb{E}[X] = \sum_{s \in S} X(s) \mathbb{P}(\{s\})$

EXAMPLE

Say the optns are A, B, C, D, and A is the correct ans for both questions.

Then, $S = \{A, B, C, D\}^2$, each $s \in S$ with prob $\frac{1}{16}$

s	$X(s)$
(A, A)	2
(A, B)	1
(A, C)	1
(A, D)	1
(B, A)	1
(B, B)	0
(B, C)	0
(B, D)	0
(C, A)	1
(C, B)	0
(C, C)	0
(C, D)	0
(D, A)	1
(D, B)	0
(D, C)	0
(D, D)	0

$\sum_{s \in S} X(s) \mathbb{P}(\{s\}) = \frac{8}{16} = \frac{1}{2}$

PROOF OF CLAIM

By def, $\mathbb{E}[X] = \sum_{x_i} x_i \mathbb{P}(X = x_i)$

$$= \sum_{x_i} x_i \mathbb{P}(\{s : X(s) = x_i\})$$

$$= \sum_{x_i} x_i \mathbb{P}\Big(\bigsqcup_{s \in S, X(s) = x_i} s\Big)$$

$$= \sum_{x_i} x_i \sum_{s \in S, X(s) = x_i} \mathbb{P}(s)$$

$$= \sum_{s \in S} X(s) \mathbb{P}(\{s\}) \quad \square$$

EXPECTED VALUE OF FUNCTIONS OF RANDOM VARIABLES

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SETTING

Let $(S, \mathbb{P})$ be a prob space
Let $X: S \to \mathbb{R}$ be a random variable
Let $g: \mathbb{R} \to \mathbb{R}$ be a function

Then, $g(X): S \to \mathbb{R}$ is another rand var

CLAIM

$\mathbb{E}[g(X)] = \sum_{x_i} g(x_i)\mathbb{P}(X = x_i)$

Proof

By the earlier claim, $\mathbb{E}[g(X)] = \sum_{s \in S} g(X(s))\mathbb{P}(\{s\}) = \sum_{x_i} \sum_{s \in S, X(s) = x_i} g(X(s))\mathbb{P}(\{s\}) = \sum_{x_i} g(x_i) \sum_{s \in S, X(s) = x_i} \mathbb{P}(\{s\}) = \sum_{x_i} g(x_i)\mathbb{P}(\bigsqcup_{s \in S, X(s) = x_i} \{s\}) = \sum_{x_i} g(x_i)\mathbb{P}(X = x_i)$ □

PROPOSITION (LINEARITY OF EXPECTATION)

Let $X_1, X_2, \ldots, X_n : S \to \mathbb{R}$ for random variables in a prob space, and let $\alpha_1, \alpha_2, \ldots, \alpha_n \in \mathbb{R}$
Then, if $X = \sum_{i=1}^{n} \alpha_i X_i : S \to \mathbb{R}$, we have $\mathbb{E}[X] = \mathbb{E}[\sum_{i=1}^{n} \alpha_i X_i] = \sum_{i=1}^{n} \alpha_i \mathbb{E}[X_i]$
(Note: Linearity of Expectation DOES NOT NEED INDEPENDENCE)

DEFINITION

Random variables $X_1, X_2, X_3, \ldots$ are independent if $\forall X_1, X_2, X_3, \ldots$, $\exists$ events $E_i := \{X = x_i\}$: indep

PROOF OF PROPOSITION

By earlier claim,
$$\begin{aligned}
\mathbb{E}[X] &= \sum_{s \in S} X(s)\mathbb{P}(\{s\}) \\
&= \sum_{s \in S} \left(\sum_{i=1}^{n} \alpha_i X_i(s)\right)\mathbb{P}(\{s\}) \\
&= \sum_{s \in S} \sum_{i=1}^{n} \alpha_i X_i(s)\mathbb{P}(\{s\}) \\
&= \sum_{i=1}^{n} \alpha_i \sum_{s \in S} X_i(s)\mathbb{P}(\{s\}) \\
&= \sum_{i=1}^{n} \alpha_i \mathbb{E}[X_i] \quad \square \text{ (by def)}
\end{aligned}$$

EXAMPLE

MCQ: n Qs, each k options
If we guess unifly rand, indep, what is the expected score?

Solution 1

Let $X :=$ # correct ans, then X takes values $\{0, 1, 2, \ldots, n\}$
We see, $\forall 0 \le r \le n$, $\mathbb{P}(X = r) = \underbrace{\binom{n}{r}}_{\text{\# ways choose correct}} \underbrace{\left(\frac{1}{k}\right)^r}_{\mathbb{P}(r \text{ corr})} \underbrace{\left(\frac{k-1}{k}\right)^{n-r}}_{\mathbb{P}(n-r \text{ wrong})}$

$\therefore$ By def, $\mathbb{E}[X] = \sum_{r=0}^{n} r\mathbb{P}(X = r) = \sum_{r=1}^{n} r\binom{n}{r}\left(\frac{1}{k}\right)^r\left(\frac{k-1}{k}\right)^{n-r}$

Solution 2

Consider the "indicator random variable" for the event of getting question i correct

$$X_i = \begin{cases} 1, & \text{ques } i \text{ corr} \\ 0, & \text{ques } i \text{ incorr} \end{cases}$$

We see: $\mathbb{E}[X_i] = 0\mathbb{P}(X_i = 0) + 1\mathbb{P}(X_i = 1) = \frac{1}{k}$

Then, # corr ans $X = \sum_{i=1}^{n} X_i$
By linearity of expectation, $\mathbb{E}[X] = \mathbb{E}[\sum_{i=1}^{n} X_i] = \sum_{i=1}^{n} \mathbb{E}[X_i] = \sum_{i=1}^{n} \frac{1}{k} = \frac{n}{k}$

Moreover, Sol 1 + Sol 2 $\Rightarrow \sum_{r=1}^{n} r\binom{n}{r}\left(\frac{1}{k}\right)^r\left(\frac{k-1}{k}\right)^{n-r} = \frac{n}{k}$

EXAMPLE

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Three investment opportunities:

Optn 1: Guaranteed \$1 a year

Optn 2: Get \$1000 w/p $\frac{1}{1000}$, \$0 w/p $\frac{999}{1000}$

with prob

Optn 3: We get \$$\frac{2000}{999}$ w/p $\frac{999}{1000}$, \$-1000 w/p $\frac{1}{1000}$

What optn should you go for?